

A comparative analysis of four classification algorithms for university students performance detection

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Abstract.

The student's performance plays an important role in producing the best quality graduate who will responsible for the country's economic growth and social development. Labor market also concern with student's performance because the fresh graduate students are considered as an employee depends on their academic performance. So, identification the reason behind student's performance variation provides a valuable information for planning education and policies. Many researchers try to find out the reason with different types of data mining approaches in different countries. But none of them worked with Bangladeshi students. This paper proposed a model for identifying the key factors of variation Bangladeshi students' academic performance and predicts their results. This paper proposes a model which able to identify the students who need special attention. Different types of feature selection methods were used such as Co-relation, Chi-Square and Euclidean distance to select valuable features. And showing the comparison of feature selections result through decision tree, Naive Bayes, K-nearest neighbor and Artificial Neural Network classifiers algorithm. The performance analysis is done by using student SGPA and review on given facilities from a university. From the performance analysis result it is found that, decreasing number of classes in dataset, the Artificial Neural Network(ANN) (93.70%) performs better than Decision Tree(DT)(92.18%), K-Nearest Neighbors (KNN)(77.74%) and Naïve Bayes (NB)(68.33%). However, increasing number of classes in dataset the DT perform better then ANN, KNN, NB.

Keywords: Educational data mining(EDM), Academic Performance(AP), Feature Selection(FS), Cumulative Grade Point Average(CGPA), Semester Grade Point Average(SGPA)

1 Introduction

Extracting knowledge from a large number of data has an impact on decision making and it has become an important component in many sectors. Educational data mining is such an emerging area through which student's learning process, learning character-

istics and academic performance can be evaluated. Educational institutions like schools, colleges and universities can use their large amount of unique data to predict the performance of their students and explore the data to better understand the learning environment. In this research, concentration been given on feature selection techniques because it is an essential part of determining what influences the performance of a data mining model. The main goal of using feature selection for this research is to find the factors of a university which have much influence on student performance. Result of feature selection shows that university admission policy, admission procedure, sincerity and commitment, laboratories facilities, internet speed, gymnasium facilities etc. along with student result have more influence in student CGPA. After identifying all those attribute predict student performance using four different algorithms, KNN, NB, DT and ANN. The performance shows that the prediction accuracy increase as the number of prediction class decrease.

This paper is organized in several sections. Section 1 provides an overview on the background of the study, problem statement and motivation, literature review and novelty of this research is presented in section 2. Section 3 provides the experimental results and Section 4 discuss on the major findings based on the experimental result. Finally, this work is concluded in section 5 with some future indications.

2 Literature Review

In recent times, data mining gaining more and more popularity of educational research field because of its potentials to educational institutes.

Data mining can be used in educational field to enhance our understanding of learning process to focus on identifying, extracting and evaluating variables related to the learning process of students as described by Alaa el-Halees [1]. Mining in educational environment is called Educational Data Mining.

Pandey and pal [2] conducted a study on the student performance based by selecting 600 students from different colleges of Dr. R.M.L Awadh University, Faizabad, India. By means of Bayes Classification on category, language and background qualification, it was found that whether new comer students will performer or not.

Hijazi and Naqvi [3] conducted a study on the student performance by selecting a sample of 300 students (225 males, 75 females) from a group of colleges affiliated to Punjab university of Pakistan. The hypothesis that was stated as “Student’s attitude towards attendance in class, hours spent in study on daily basis after college, student’s family income, student’s mother’s age and mother’s education are significantly related with student performance” was farmed. By means of simple linear regression analysis, it was found that the factors like mother’s education and student’s family income were highly correlated with the student academic performance.

Baradwaj and Pal [4] collected data like attendance, class test, seminar and assignments marks from university student’s previous database and predict the performance at the end of the semester.

Al-Radaideh, et al [5] applied a decision tree model to predict the final grade of students who studied the C++ course. He was used three different classification meth-

ods namely ID3, C4.5, and the NaiveBayes. After calculation the results indicated that Decision Tree model had better prediction than other models.

Galit [6] uses student's data to give a case study which analyzes students learning behavior to predict the results and to warn students at risk before final exams.

Bhardwaj and Pal [7] selected 300 students based on their academic performance from 5 different degree colleges who conducting Bachelor of Computer Application Course. By using the classification method, the result indicated that the factors like student's grade in the senior secondary exam, living location, the medium of teaching, mother's qualification, student's other habit, family annual income, and student's family status were highly correlated with students' academic performance.

Nawal Ali Yassein et al. [8] have proposed a patterns to predict students' performance. The research involves two parts, firstly find out the factors that are related to course success rate, and another one to determine predictors based on students performance. Both classification and clustering techniques were used to find out the factors that are affect students' performance.

Raheela Asif et al [9] proposed a case study based on predicting student academic performance. By using different classifiers, data have been extracted from 347 undergraduate students. Artificial neural networks, decision trees, k-nearest Neighbor, naive Bayes, rule induction classification techniques have been used in a study. The study has shown the possibility of predicting fourth-year graduation performance at university with pre-university marks and initial university two-year marks. The accuracy on analyzing the datasets was satisfactory.

Nguyen Thai Nghe, Paul Janecek, Peter Haddawy [10] predicts academic performance considering the data of two different universities. Firstly, they collected and use the data of undergraduate students from Can Tho University (CTU) in Vietnam to predict GPA by using the student's record (faculty, English skill, age, gender, family, job, religion etc) and 2nd year GPA. Secondly, they consider the data from masters' students of Asian Institute of Technology (AIT). By using students' admission information (like entry GPA, English proficiency marital status, academic institute, TOFEL score, Income age etc.) they predict the GPA of students at the end of 1st year of the masters' degree. Decision tree and Bayesian networks have been used in this study and accuracies of these algorithms are also compared. The accuracies are as follows: 73/71% for {fail, fair, good, very good}, 84/67% for {fail, good and very good} and 94/93% for {fail, pass} at the CTU/AIT respectively. the prediction the accuracy of 2 classes is much higher than that of 3 classes or 4 classes. They obtain higher accuracies using the decision tree classifier.

Antonia Lozano Díaz [13] collected data from secondary schools in Almeria City(Spain). She was composed of total 1178 students, including the four years of mandatory secondary education. An adaption of the TAMAI questionnaire and a measurement of school failure have been used. The results of this survey make clear the direct influence of variables such as parents' academic level, gender, motivation, relationships between peers, and others.

Amirah Mohamed Shahin, Wahidah Husain, Nur'aini[15] predicted students performance using datamining. They focuses on how the prediction algorithm can be used to identify the most important attributes in a students data.

Asif, Raheela, Agathe Merceron, Syed Abbas Ali, and Najmi Ghani Haider[16] predicted students' academic achievement at the end of a four year study program and identified the low and high achieving students which could result in timely warning and support a low achieving students.

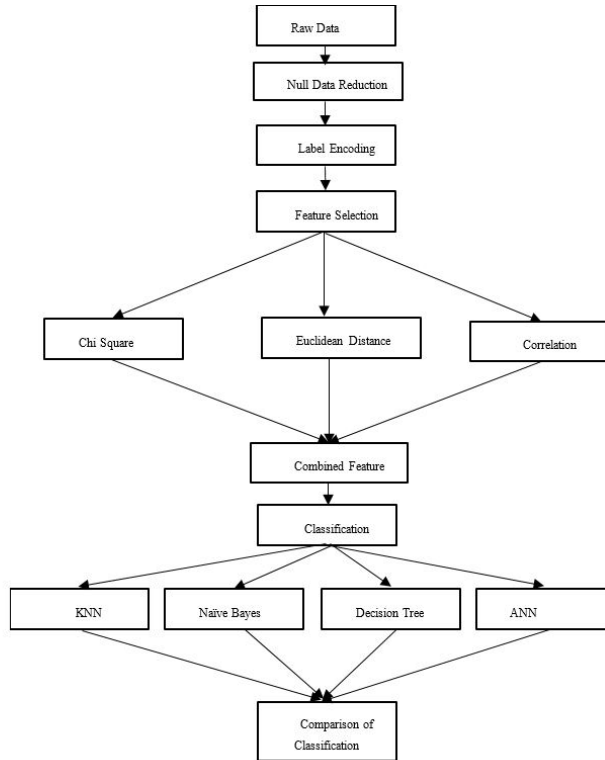
Fok, Wilton WT, Y. S. He, HH Au Yeung, K. Y. Law, K. H. Cheung, Y. Y. Ai, and P. Ho[17]

With thousand students' data implemented artificial intelligence and deep learning algorithm provided another approach for intelligent classification and result prediction.

These work shows that it is possible to predict student performance based on one background qualification, attendance, class test result, assignments mark, English skill, age, gender, family status, religion, marital status, attitude toward class, daily hours spent on study, family income, mother qualification, mother age, living location, living cost of student etc. Researcher also predict one's performance using a certain subject one studied. Most of the case analyzer use classification and clustering technique Where in few case they tried to find factors that are related to course. Now days more data are generated by university while they take review from student. Where comes a question, is it possible to predict one performance using students review and other available data?

3 Methodology

This section of the paper describe how data been collected and analysis of the student academic performance is done. Different task that have been done throughout this research is described here.



3.1 Data Collection

Precise data collection is fundamental to keeping up the trustworthiness of research, settling on educated business choices and guaranteeing quality affirmation. Data collection methods can be divided into two categories: secondary methods of data collection and primary methods of data collection. We used here both categories. Firstly, in secondary data collection methods, for the research purpose, the dataset gathered from web source named "Kaggle" (Student Survey, University level student survey for academic quality enhancement, 2017) [14]. We download a dataset concentrated the survey analysis on the students' quality enhancement. We named the dataset as Student Survey(SS). This dataset was collected by a survey conducted by Google forms for a Bangladeshi University in order to examine their current academic situation and also to improve on them. This Survey was part of the Institutional Quality Assurance Program, initiated by the University Grant Commission, Bangladesh and

funded by the World Bank. After collecting data, we calculated the average CGPA for each student. Which are then used for further evaluation.

3.2 Null Data reduction

In the SS database, there are some common properties such as incomplete, noisy and inconsistent data. There are various fields that have no recorded data for a number of rows, then the null values are removed through calculating the average of the column and putting the average value on null field. This method is not efficient when the percentage of missing values are increased. The algorithm for removing null data from dataset is shown in Figure 1.

```

1. Load Dataset ();
2. For each Column
3. {Calculate average of column;
4.  IF row of Column contains Null
5.    {Put average of column to Null field; }
6.  Else
7.    {Continue; }
8. }
9. Save_Dataset ();

```

Fig 1. Algorithm for removing null data from dataset

3.3 Label Encoding

In this research the preparation data is named by words for mapping. To change over word names into numbers, a name encoder can be used. The name encoding refers to the manner toward changing the word names into the numerical frame. The coding label of categorical value, there is a function encloses in panda library to code the data and reverse the encoded values. The algorithm for label encoding is shown in Figure 2.

```

Load_DataSet()
For each column of columns (Number of total columns/attribute) {
  unique_values ← Finding unique values of the column
  For I = 0 to M – 1 (M number of unique values) {
    Encoding ← Encoded index of I unique_values
  }
}
Save_DataSet()

```

Fig 2. Algorithm for label encoding

3.4 Feature Selection

Chi Square

The Chi-Square test of freedom is utilized to decide whether there is a significant connection between two categorical variables. The recurrence of every classification for one ostensible variable is looked at over the classifications of the second ostensible variable. From the definition, of chi-square we can easily deduce the utility of chi-square approach in feature selection. Equation for Chi square is [14]

$$\chi^2 = \sum \frac{(O-E)^2}{E} \quad (1)$$

Where O is our observed attribute like student SGPA, gender, *curriculumPressure*, *assessmentMeetsObjectives* etc. And E is the Expected value which is Student CGPA. Chi Square with null hypothesis found out 39 important attribute along with different SGPA there is *studentType*, *curriculumPressure*, *coursesCurriculumArranged*, *modernDevicesUsed*, *admissionPolicy* etc. attribute is important feature of the dataset.

Euclidean Distance

Euclidean distance is usually used for distance. Euclidean distance examines the core of the square difference within the pair of objects. For each feature, Euclidean distance calculate from it to all other features. Euclidean distance calculated features between and using the formula: [14]

$$(x, y) = \{\sum i(X_i - Y_i)^2\}^{\frac{1}{2}} \quad (2)$$

Euclidean distance found that along with SGPA, *internetSpeed*, *indoorAndOutdoor medicalFacilities* have less distance with the output result. As 38 important attribute selected through chi square there 39 attribute with lowest distance selected for further evaluation.

Correlation

Correlation algorithm find that, the correlation between input attribute and output attribute. On our dataset *libraryMaterials*, *resultAssessment*, *classroomFacilities*, *internetSpeed*, *indoorandoutdoorMedicalMacilities* among with SGPA seem to be more correlated with student academic performance and then top 39 correlated attribute been selected for further evaluation [14].

$$r_x = \frac{k\bar{y}_{z1}}{\sqrt{k+k(k-1)\bar{r}_{11}}} \quad (3)$$

Combining Feature Selection Result

From Chi square we found 39 feature as important. Then taking most relevant 39 from other feature selection algorithm. For combining Feature, we selected the feature that are within best 39 result of each feature selection algorithm. From the best 39 feature 22 common feature are selected and named combined features: Such as, *1stY1stSem*, *1stY2ndSem*, *1stY3rdSem*, *2ndY1stSem*, *2ndY2ndSem*, *2ndY3rdSem*, *3rdY1stSem*, *3rdY2ndSem*, *3rdY3rdSem*, *4thY1stSem*, *4tyY2ndSem*, *admissionPolicy*, *admissionProcedure*, *sincerityAndCommitment*, *laboratoriesFacilities*, *internetSpeed*, *gymnasiumFacilities*, *adequateSafety*, *scholarshipsAvailable*, *oppertunitiesOfcommunityServices*, *departmentDevelopmentPolicy* and *departmentServicePolicy*.

4 Results and Discussion

The Prediction Summary on dataset using DT, KNN, NB and ANN is shown on Figure 3 and Figure 4. These graph show details result for 3, 5 and 9 class with different test and train data. Here 3 class is used as considering CGPA less than 2.00 as class 0, CGPA within 2.00-3.24 as class 1 and CGPA 3.25-4 as class 2. 5 class is used as considering CGPA less than 2.00 as class 0, 2.00-2.49 as class 1, 2.50 to 2.99 as class 2, 3.00-3.49 as class 3 and 3.50 to 4.00 as class 4. 9 class is used as considering CGPA less than 2.00 as 0, 2.00-2.24 as class 1, 2.25-2.49 as class 2, 2.50-2.74 as class 3, 2.75-2.99 as class 4, 3.00-3.24 as class 5, 3.25-3.49 as class 6, 3.50-3.74 as class 7, 3.75-4 as class 9. Result shows that as the number of class decreased accuracy improved in significant number. DT perform better with large number of class but with feature selection and less class ANN Gradually improve and beat DT. The accuracy of prediction after feature selection significantly improved.

Figure 3 shows that KNN Perform lower than other three algorithms where Naive Bayes(NB) slight better KNN with less train data but KNN perform just Better then NB with more train data. Where NB got average accuracy of 22.91% with 9 class and KNN got 22.93%. Where ANN does comparatively better than KNN and NB with 36.12% accuracy with 9 class and 76.50% with 3 class. But with More class DT Definitely outperform KNN, NB, and ANN with better accuracy of 59.20% with class 9 And as the number of class decrease such as for 3 class DT average accuracy is 90.91%.

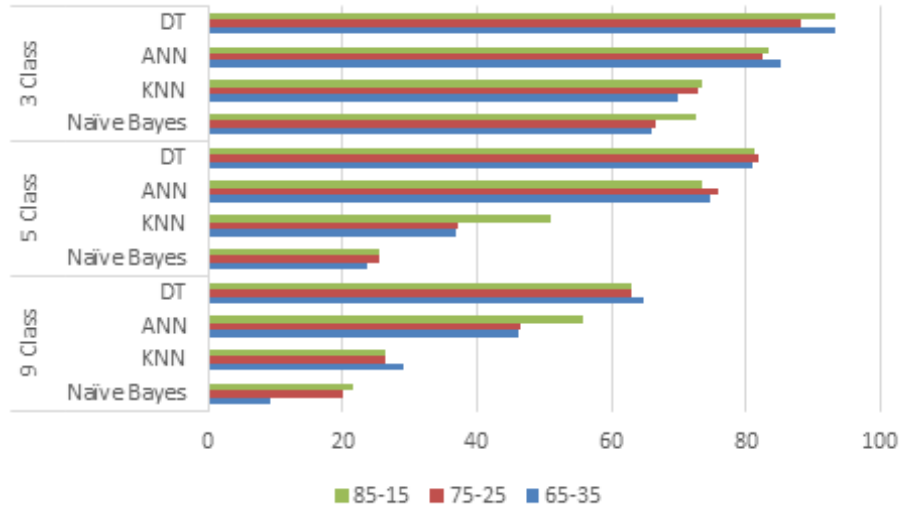


Fig. 3. Comparison of student performance prediction percentage before feature selection

Figure 4. shows that the combined feature selection results. It is found from Figure 4, the increasing number of class decreasing the accuracy of KNN and NB algorithm result, however the decreasing number of class improve KNN and NB accuracy. DT here also perform well with the large amount of class but as the classifier decrease to 3 from 9 ANN come with better performance with average accuracy of 93.70%. Where DT have accuracy which is 92.18% close to ANN. Therefore, lower number of class selection may tend to the more accurate result. Figure 2 shows that highest accuracies are achieved by using 3 class. Hence, depending on the selection of features is a crucial fact to get better accuracy for predicting student academic performance.

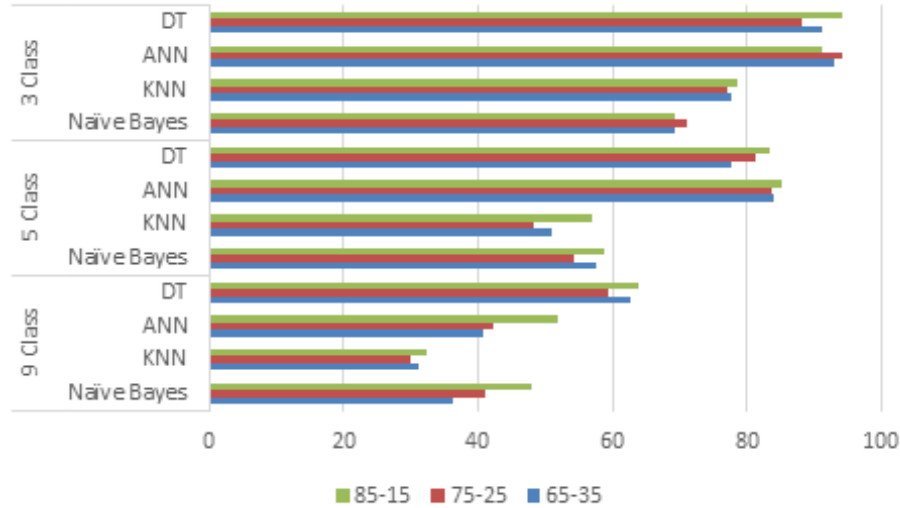


Fig. 4. Comparison of student performance prediction percentage after feature selection

5 Conclusions and Future Directions

In this research Data mining techniques such as, feature selection and classification are applied on SS dataset [14] and predict university students performance. Three feature selection algorithms (such as, Chi-square, Euclidean distance and correlation) results are combined for predicting university students' performance. The KNN, NB, DT and ANN classification algorithms are used to predict student performance using 9 class, 5 class and 3 class. DT perform the best with accuracy of 60.35% for 9 class. Even for 5 class DT perform better with 11.83% more accuracy then ANN. But with 3 class ANN perform 1% more accurately then DT. Therefore, it can be concluded that increasing number of class may lead to get less accuracy; however, decreasing number of class may increase the accuracy. This research can be extended different ways. Such as, more feature selection algorithm can be used, more classification algorithm can be used to evaluate the result more accurately. This research paper predicts student performance as CGPA using student SGPA and their review on university facilities, on further work student SGPA can be predict using this dataset.

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